

The X/P Asymmetry Was Never a Law: A Four-Way Split at Four Grade Levels

Harald Rønning

Question. A companion note shows that among generalized Inönü–Wigner contractions of the imaginary octonions sharing the Günaydin–Lüst–Malek $1+3+3$ grading shape, genuine (non-associative) R-flux arises if and only if the momentum subspace, not the coordinate subspace, is the one that closes into a subalgebra on its own — an exact, exhaustively verified asymmetry, but explicitly scoped to that one shape. Does this asymmetry reflect something structural about which side of a grading is "allowed" to carry genuine content, or was it an artifact of that shape specifically having no room for the other side's self-bracket to ever survive?

Answer. The latter. Two natural 4-level shapes turn out to be combinatorially impossible outright — one provably, in general, by a clean Fano-plane argument. A third does exist, and once tested exhaustively, the clean asymmetry dissolves into a four-way split: either side can be the one required to stay open, or both can carry genuine content simultaneously. The original rule was never a law about coordinates versus momenta — it was a direct consequence of the $1+3+3$ shape leaving nowhere for the momentum self-bracket to land, so its closure was free regardless of any deeper principle.

1. Background

The companion result: in the $1+3+3$, grade- $(1, 2, 3)$ family, $\{X, X\}$ (grade $1+1=2$) has a real target — grade 2, occupied by P — while $\{P, P\}$ (grade $2+2=4$) does not, since no grade-4 level exists in a 3-level system. Consequently P 's closure never costs anything: the bracket it would kill was already forced to vanish by the shape itself. This section tests whether the same asymmetry survives once a shape actually provides both self-brackets somewhere to land.

2. Two shapes, ruled out first

Attempt 1: grades 1, 2, 3, 4, sizes 3, 2, 1, 1 (X, P, Q, W). Exhaustive search over all 420 possible assignments of the seven octonion generators to this shape found zero satisfying the Weimar-Woods validity condition. The mechanism was traced by direct inspection for representative cases (every failure involved an X - P pair, grade 3, landing on W , grade 4) but no fully general combinatorial proof of impossibility was derived for this specific shape; it is reported here as a verified computational fact.

Attempt 2: grades 1, 2, 4 (skipping 3), sizes 3, 3, 1 (X, P, W). Also exhaustively found impossible — 140 assignments, zero valid. Here a clean, general proof is available.

Lemma (Fano matching obstruction). For any point w among the seven imaginary octonion units, the three unordered pairs $\{a, b\}$ satisfying $a \times b = \pm w$ partition the remaining six points into a perfect matching.

Proof. w lies on exactly three of the Fano plane's seven lines. Each such line contains w and two other points, whose product is $\pm w$. Any two distinct lines of the Fano plane meet in exactly one point; since both lines here pass through w , they share no other point. So the "other two points" on each of the three lines through w are pairwise disjoint, giving three disjoint pairs — six points total, exactly the six remaining. ■

Corollary. No assignment splitting those six points into two groups of three, with all cross-pairs required to avoid w , can exist. Each pair needs both its elements in the same group to avoid being a violating cross-pair; a group of size 3 can hold at most one pair intact (using two of its three slots); two groups can jointly hold at most two of the three pairs intact; the third pair is unavoidably split, and a split pair is exactly a cross-pair landing on w . ■

This directly rules out Attempt 2 for every choice of w , not merely the one tested.

3. A valid 4-level shape exists

An exhaustive search over grade-assignment functions $\{1, \dots, 7\} \rightarrow \{1, \dots, 7\}$, filtered to those using at least four distinct values, found valid examples immediately. A representative: groups $A = \{1, 2\}$, $B = \{3, 4\}$, $C = \{5, 6\}$ (grades 1, 2, 3) and $D = \{7\}$ (grade 4) — sizes $2+2+2+1$. All 21 pairwise products were checked directly against the Weimar-Woods condition and found valid without exception. Critically, $[e_3, e_4] \rightarrow e_7$: B 's self-bracket (grade $2+2=4$) lands exactly at D 's grade 4 — the condition that could never be satisfied in the $1+3+3$ shape now genuinely holds.

4. The four-way split

All 420 valid (partition, grade-assignment) combinations sharing this $2+2+2+1$ size profile were enumerated, and for each, whether the grade-1 and grade-2 groups are self-closed, and whether their self-brackets survive contraction (land exactly at $2\times$ their own grade), was checked directly.

grade-1 closed	grade-1 survives	grade-2 closed	grade-2 survives	count
No	Yes	No	Yes	84
No	Yes	Yes	No	126
Yes	No	No	No	126
Yes	No	No	Yes	84

Row 2 reproduces the companion paper's original pattern exactly (lower grade open and genuine, higher grade closed and trivial). Row 4 is its mirror: the *lower* grade closed and trivial, the *higher* grade open and genuine — a direct reversal, impossible in the $1+3+3$ shape by construction. Row 1 shows both self-brackets genuinely surviving *simultaneously*, something the earlier shape could never produce regardless of any

grouping choice, since grade 4 simply did not exist. Row 3 shows that openness alone is not sufficient — a bracket can exit its own span without landing at the specific grade needed to survive contraction.

5. Why the earlier asymmetry looked like a law

In the $1+3+3$ shape, $\{P, P\}$'s target grade (4) exceeded the shape's maximum (3) unconditionally — closure or not, that bracket was always going to vanish. The observed asymmetry was not evidence that momentum-type closure is *inherently* harmless while coordinate-type closure is *inherently* costly; it was evidence that one side had a real target to protect and the other side's fate was already sealed by the shape before any grouping choice was made. Once a shape provides a real target on both sides — as $2+2+2+1$ does — the asymmetry dissolves into the genuine four-way structure of §4, and either side can play either role.

6. Scope

The mechanism behind Attempt 1's impossibility (§2) was verified computationally but not reduced to a general proof the way Attempt 2's was; whether a comparable Fano-theoretic argument exists is open. The exact counts in §4 (84 vs. 126) sum consistently ($210 + 210 = 420$, an even split by whether the grade-1 group is closed) but their precise combinatorial origin was not derived here. Whether shapes with five or more grade levels produce further, qualitatively new regimes beyond this four-way split is untested.

References

- Companion note, "Which Contractions of the Octonions Give Genuine R-Flux: A Necessary and Sufficient Condition."
- Günaydin, M., Lüst, D. & Malek, E., "Non-associativity in non-geometric string and M-theory backgrounds, the algebra of octonions, and missing momentum modes," arXiv:1607.06474.

Created in collaboration with Claude (Anthropic). All conclusions and any errors remain the author's responsibility.